

Robotic Budding: From Homogeneous Cells to Diverse Morphologies in Physical Self-Reproduction



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01 Introduction

Core Question: Can a single homogeneous module support both diverse body plans and physical self-reproduction?

Key Approach: Budding-style physical reproduction where a parent robot sequentially collects loose modules to build an identical copy.

Contributions:

- Homogeneous module design with magnetic docking.
- Demonstrations across two distinct morphologies (3-module Trilobite & 5-module Seal).

02 Reproduction Formulation

Module Sets

Parent: $\mathcal{O} = \{o_1, \dots, o_n\}$ (initial modules)

Environment: $\mathcal{E} = \{e_1, \dots, e_n\}$ (scavenged modules)

Reproduction Workflow

$$R(\mathcal{O}) + \mathcal{E} \xrightarrow{\pi_{\text{collect}}} R(\mathcal{O} \cup \mathcal{E}) \xrightarrow{\pi_{\text{separate}}} R(\mathcal{O}) + R(\mathcal{E})$$

π_{collect} : Sequential module acquisition ($n \rightarrow 2n$)

π_{separate} : Antagonistic joint motion disconnecting target interface

Structural Equivalence & Budding Lineage

$$\text{Isomorphism: } G_p \simeq G_o \simeq G_n^*, |V(G_p)| = |V(G_o)| = n$$

$$\text{Provenance Index: } \sigma = \frac{|V(G_p) \cap \mathcal{O}| + |V(G_o) \cap \mathcal{E}|}{2n} = 1$$

$\Rightarrow$ Budding-Style Reproduction

Parent retains original hardware;
copy consists entirely of acquired modules.

03 System Overview

Each module consists of two triangular segments connected by two actuated revolute joints. The joints allow the module to fold from a planar configuration into a three-dimensional structure.

Five embedded neodymium magnets enable reversible, direction-independent docking. Each module also contains a battery, servo actuators, and an onboard microcontroller.

Target Morphologies

Trilobite: 3 modules

Seal: 5 modules

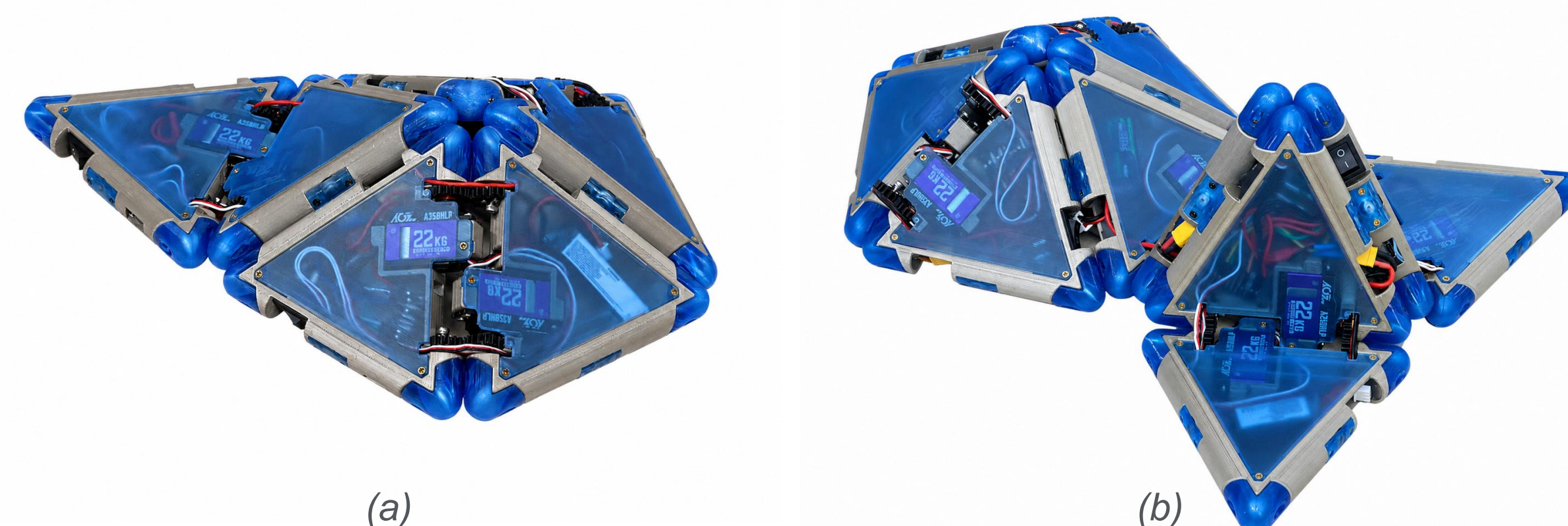


Fig. 1: (a) The assembled three-module trilobite morphology. (b) The assembled five-module seal morphology.

04 Control Architecture

Collection and assembly are currently teleoperated through a browser-based control terminal.

The operator sends three discrete commands: forward, left, and right, to a host microcontroller. Commands are broadcast wirelessly to the modules, which return periodic heartbeat signals.

Docking is confirmed through direct visual observation rather than force or current sensing.

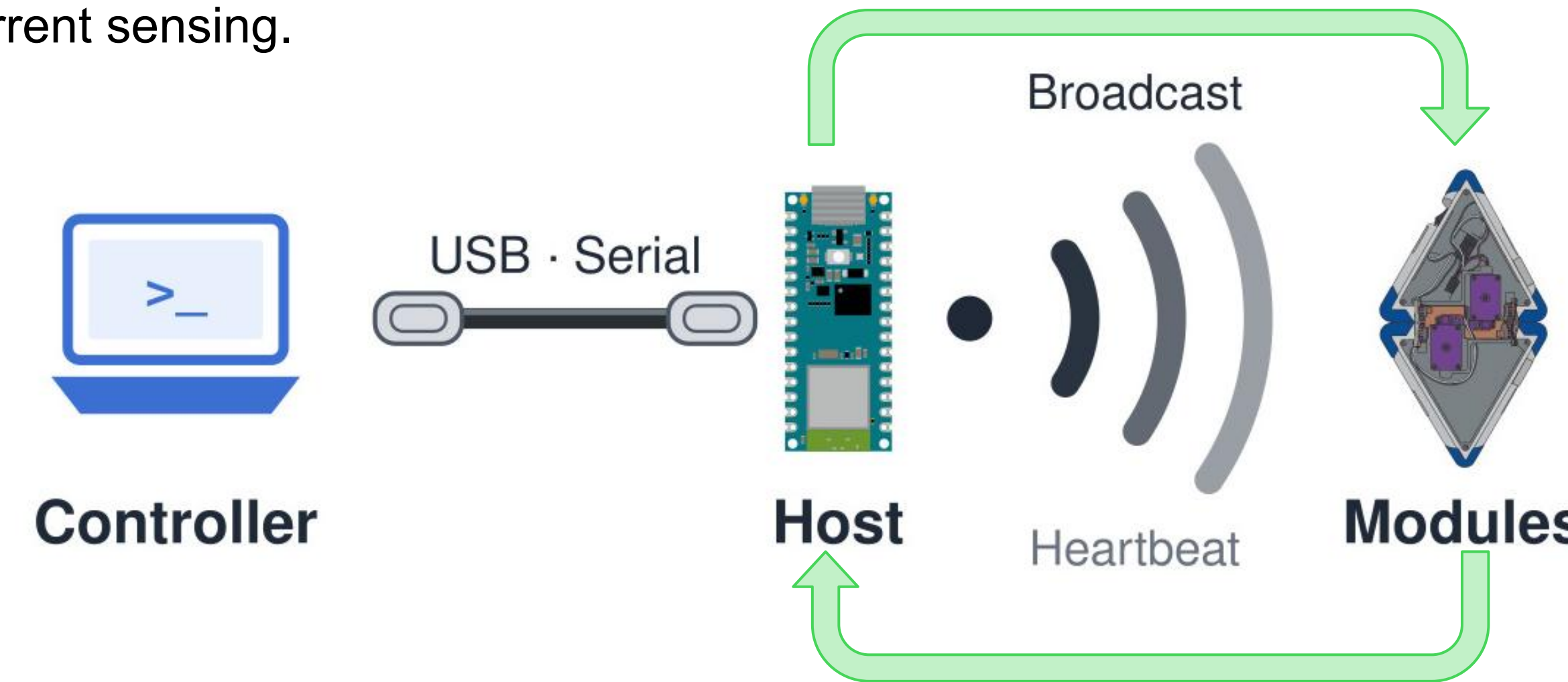


Fig. 2: Control system communication architecture.

05 Collection & Assembly

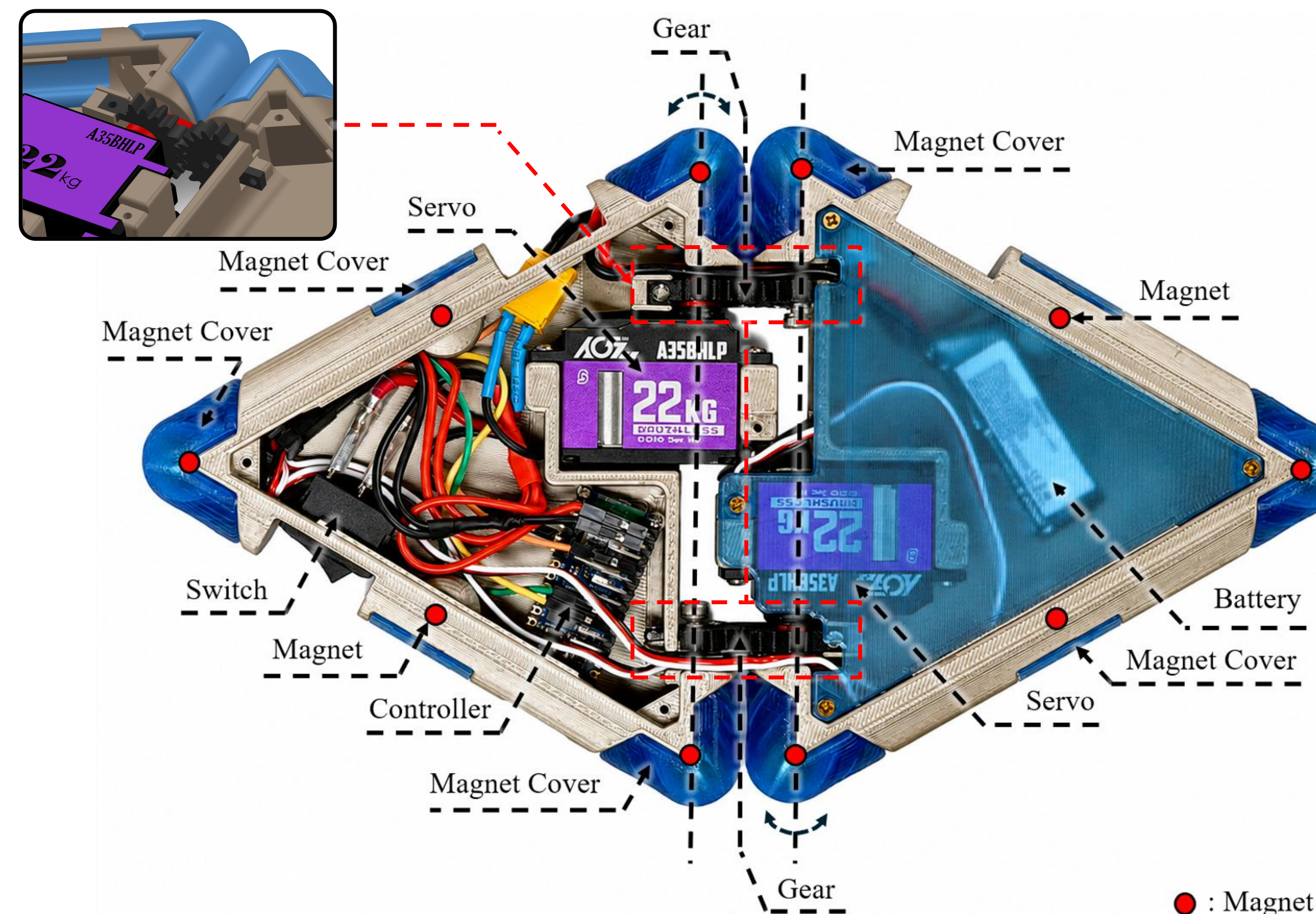
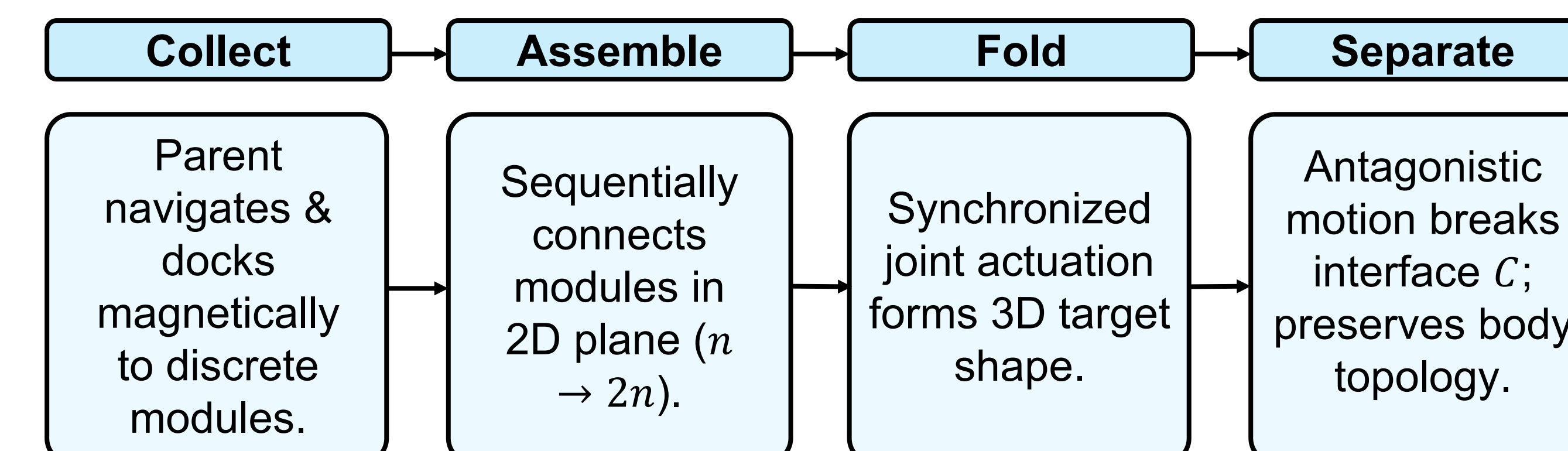


Fig. 3: Internal structural layout and component allocation of an individual module, highlighting the dual-servo drive mechanism, onboard electronics, and magnetic docking interfaces (red dots).

06 Demonstrations & Preliminary Results

Three-Module Trilobite

A trilobite parent collected three modules and assembled a structurally identical offspring.

Five-Module Seal

A seal parent collected five modules, formed a ten-module composite, and separated into two identical five-module robots.

Key Result

- Successful collection, folding, and separation
- Structural equivalence after reproduction
- Independent post-separation locomotion



1. Straight forward 2. Left forward 3. Right forward
 Fig. 4: Locomotion sequence of the five-module seal robot. Forward locomotion (1.a~1.d). Left steering (2.a~2.d). Right steering (3.a~3.d). Snapshots in each column show the deformation cycle from top to bottom ($t = a \rightarrow b$).

One successful demonstration was completed for each morphology; recursive reproduction has not yet been tested.

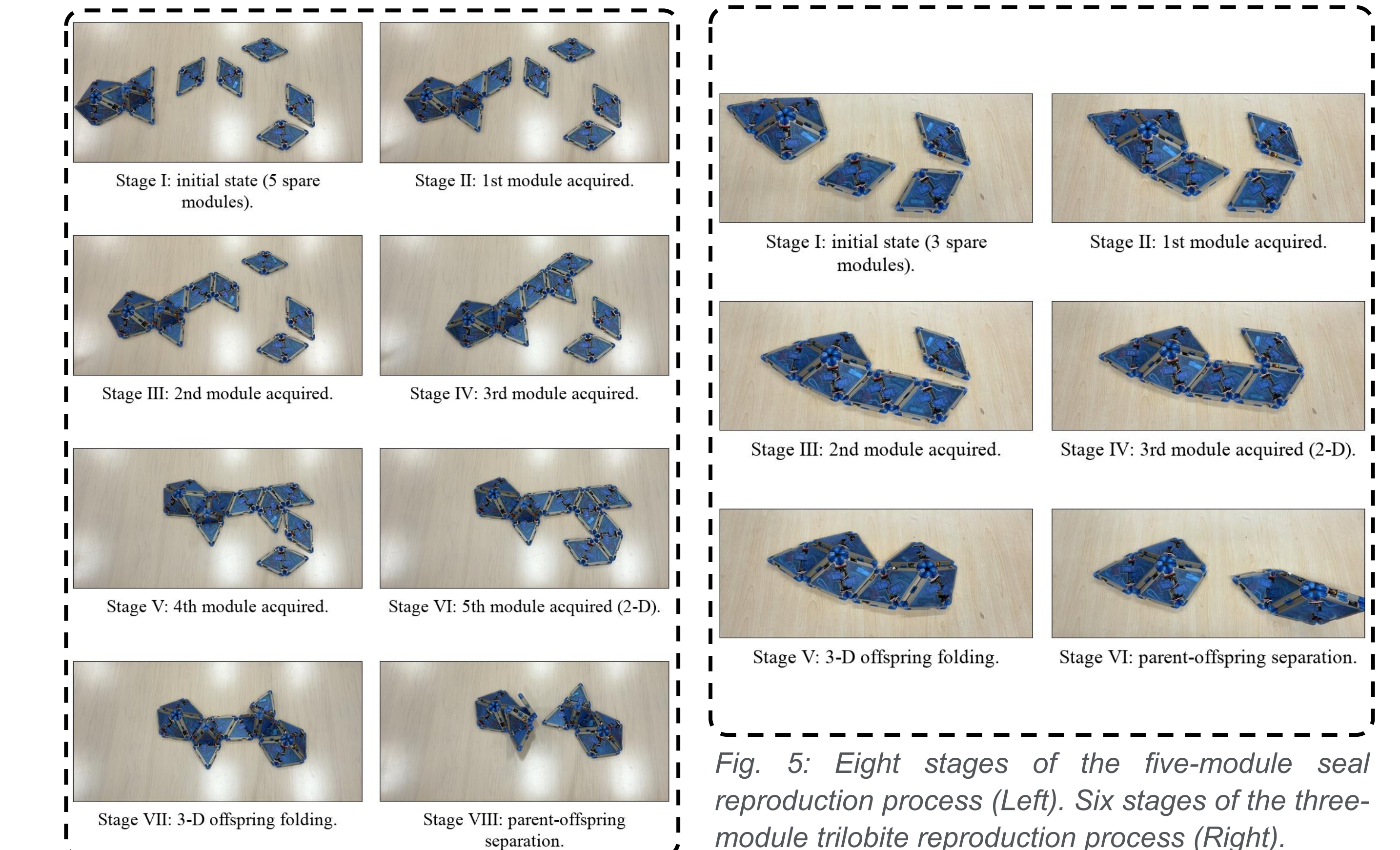


Fig. 5: Eight stages of the five-module seal reproduction process (Left). Six stages of the three-module trilobite reproduction process (Right).

07 Discussion & Future Work

Current Scope

Control: Human teleoperation via browser interface.
 Docking: open-loop visual confirmation
 Validation: Single demonstration per morphology.
 Lineage: First-generation offspring only.

Next Steps

Autonomy: Vision-based navigation & trajectory planning.
 Sensing: Closed-loop physical docking feedback.
 Robustness: Multi-trial quantitative evaluation.
 Evolution: Multi-generation recursive reproduction.

Vision: Transitioning from teleoperated budding to fully autonomous and recursive robotic reproduction.